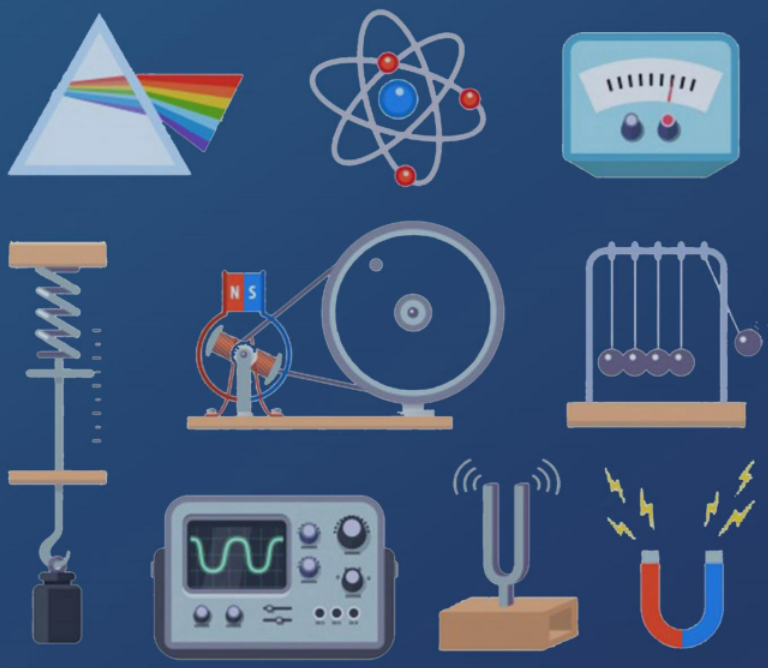
PHYSICS

Atomic Structure



SHEET

eSaral.com

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Discovery of electron and its charge	1			
2	Rutherford model and the discovery of proton,nucleus and neutron		1		
3	Size of nucleus and drawbacks of Rutherford model	2,3			
4	Wave Basics	4			
5	EM waves and Planck's theory	5	2		
6	Bohr's model	6-9	3,4,7,11-13	1-6	1-9,19
7	Spectrum and Rydberg's formula	10-14	5,14	7-10	10-14, 21,23
8	Drawback of Bohr, light: wave or particle?	15			
9	De broglie explanation of bohr's second postulate of Quantisation	16,17		11	
10	After Full Chapter		6,8,9,10		15-18, 20, 22

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Discovery of electron and its charge

Q1. Charge on electron is -
(A) -1.6×10^{-19}C (B) -1.6×10^{-10}C
(C) -4.8×10^{-10}C (D) Zero

Size of nucleus and drawbacks of Rutherford model

Q2. The size of an atom is of the order of :
(A) 10^{-8} cm (B) 10^{-10} cm
(C) 10^{-12} cm (D) 10^{-6} cm

Q3. An atom having atomic mass number 13 has 7 neutrons. What is the atomic number of the atom ?
(A) 6 (B) 7
(C) 8 (D) 9

Wave Basics

Q4. Calculate the wave number of a radiation having wavelength 600 nm ?
(A) 1.67×10^{8} m^{-1} (B) 1.67×10^{10} m^{-1}
(C) 1.67×10^{12} m^{-1} (D) 1.67×10^{6} m^{-1}

EM waves and Planck's theory

Q5. The frequency of an electromagnetic radiation is 2×10^{6} Hz. What is its wave-length in metres
(Velocity of light = 3×10^{8} ms^{-1})
(A) 6.0×10^{14} (B) 1.5×10^{4}
(C) 1.5×10^{2} (D) 6.0×10^{-2}

Bohr's model

Q6. The energy required to remove an electron in a hydrogen atom from n = 10 state is-
(A) 13.6 eV (B) 1.36 eV
(C) 0.136 eV (D) 0.0136 eV

Q7. The binding energy of an electron with n = 4 in the H-atom is-
(A) 13.6 eV (B) 3.4 eV
(C) 1.51 eV (D) 0.85 eV

Q8. The velocity of electron in the first Bohr orbit of He^{+} ions is-
(A) c (B) c/137
(C) 2c/137 (D) 4c/137

Q9. The ratio of energies of first two excited states hydrogen atom is -
(A) 3/1 (B) 1/4
(C) 4/9 (D) 9/4

Spectrum and Rydberg's formula

Q10. The number of different wavelengths may be observed in the spectrum from a hydrogen sample if the atoms are excited to third excited state is -
(A) 3 (B) 4
(C) 5 (D) 6

Q11. The transition from the state n = 4 to n = 3 in a hydrogen like atom results in ultraviolet radiation. Infrared radiation will be obtained in the transition :
(A) $2 \rightarrow 1$ (B) $3 \rightarrow 2$
(C) $4 \rightarrow 2$ (D) $5 \rightarrow 4$

Q12. The shortest wavelength present in the Brackett series of a spectral line is-
(A) 912Å (B) 8201Å
(C) 1.46 μm (D) 2.28 μm

Q13. Out of following spectral series of hydrogen spectrum, the one which lies wholly in the ultraviolet region is-
(A) Balmer series
(B) Brackett series
(C) Paschen series
(D) Lyman series

Q14. The wave number of the series limit of Lyman series is-

(A) $1.097 \times 10^7\ m^{-1}$

(B) $2.74 \times 10^6\ m^{-1}$

(C) $1.22 \times 10^6\ m^{-1}$

(D) $6.86 \times 10^5\ m^{-1}$

Drawback of Bohr, light : wave or particle?

Q15. According to the modern theory for nature of light, the light has-

(A) wave nature only

(B) particle nature only

(C) dual nature

(D) none of these

De broglie Explanation of Bohr's second postulate of quantisation

Q16. According to De Broglie, wavelength of electron in second orbit is 10^{-9} metre. Then the circumfrance of orbit is-

(A) 10^{-9} m (B) 2×10^{-9} m

(C) 3×10^{-9} m (D) 4×10^{-9} m

Q17. When a hydrogen atom emits a photon of energy 12.1 eV,its orbital angular momentum changes by :

(A) $1.05\times10^{-34}\,J-s$

(B) $2.11\times10^{-34}\,J-s$

(C) $3.16\times10^{-34}\,J-s$

(D) $4.22\times10^{-34}\,J-s$

Exercise - 1A Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Rutherford's experiment established that :

(A) Inside the atom there is a heavy positive centre

(B) Nucleus contains protons and neutrons

(C) Most of the space in the atoms is empty

(D) Size of the nucleus is very small

Q2. The energies E_1 and E_2 of two radiations are 25 eV and 50eV respectively. The relation between their wavelengths i.e. λ_1 and λ_2 will be?

(A) $\lambda_1 = \lambda_2$ (B) $\lambda_1 = 2\lambda_2$

(C) $\lambda_1 = 4\lambda_2$ (D) $\lambda_1 = \frac{1}{2}\lambda_2$

Q3. For a doubly ionised Li–atom

(A) Angular momentum of an electron in 3rd orbit is $\frac{3h}{2\pi}$

(B) Energy of electron in 2nd excited state is –13.6 eV

(C) Speed of electron in 3rd orbit is $\frac{c}{137}$, where c is speed of light

(D) Kinetic energy of electron is 2nd excited state is half of the magnitude of the potential energy

Q4. A particular hydrogen like atom has its ground state binding energy 122. 4 eV. It is in ground state. Then -

(A) Its atomic number is 3.

(B) An electron of 90 eV can excite it.

(C) An electron of kinetic energy nearly 91.8 eV can be brought to almost rest by this atom.

(D) An electron of kinetic energy 2.6 eV may emerge from the atom when electron of kinetic energy 125 eV collides with this atom.

Q5. An electron in hydrogen atom after absorbing energy photons can jump between energy states n_1 and n_2 ($n_2 > n_1$). Then it may return to ground state after emitting six difference wavelengths in emission spectrum. The energy of emitted photons is either equal to, less than or greater than the absorbed photons. Then n_1 and n_2 are :-

(A) $n_2 = 4, n_1 = 3$

(B) $n_2 = 5, n_1 = 3$

(C) $n_2 = 4, n_1 = 2$

(D) $n_2 = 5, n_1 = 1$

Q6. A beam of ultraviolet light of all wavelengths passes through hydrogen gas at room temperature, in the x–direction. Assume that all photons emitted due to electron transition inside the gas emerge in the y–direction. Let A and B denote the lights emerging from the gas in the x and y directions respectively.

(A) Some of the incident wavelengths will be absent in A.

(B) Only those wavelengths will be present in B which are absent in A.

(C) B will contain some visible light.

(D) B will contain some infrared light.

Q7. The magnitude of angular momentum, orbit radius and frequency of revolution of electron in hydrogen atom corresponding to quantum number n are L, r and f respectively. Then according to Bohr's theory of hydrogen atom

(A) fr^2L is constant for all orbits

(B) frL is constant for all orbits

(C) f^2rL is constant for all orbits

(D) frL^2 is constant for all orbits

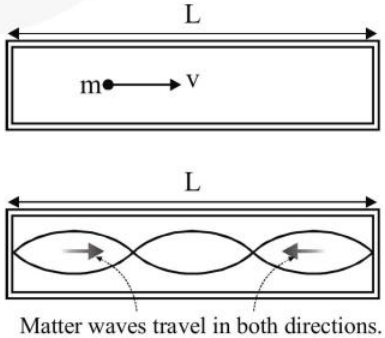
PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question. 8 to 10

While conducting his doctoral research in theoretical physics and with no experimental evidence to go on, de Broglie reasoned by analogy with Einstein's equation E = hf and with some of the ideas of his theory of relativity. The details need not concern us, but they led de Broglie to postulate that if a material particle of momentum p = mv has a wave-like nature, then its wavelength must be given by $\lambda = \frac{h}{p} = \frac{h}{mv}$ where h is Planck's constant. This is called the de-Broglie wavelength.

de-Broglie considered a matter wave to be a traveling wave. But suppose that a "particle" of matter is confined to a small region of space and cannot travel. How do the wave-like properties manifest themselves? This is the problem of "a particle in a box." Figure shows a particle of mass m moving in one dimension as it bounces back and forth with speed v between the ends of a box of length L. We'll call this a one-dimensional box; its width isn't relevant. A particle in a box creates a standing de Broglie wave as it reflects back and forth.



Matter waves travel in both directions.

Q8. What should be de-Broglie wavelength of confined particle in the box [here n∈N].

(A) $\frac{L}{2n}$ (B) $\frac{2L}{n}$ (C) $\frac{L}{n}$ (D) nL

Q9. Confined particle's energy is given by :

(A) $\frac{n^2h^2}{2mL^2}$ (B) $\frac{2n^2h^2}{mL^2}$

(C) $\frac{n^2h^2}{8mL^2}$ (D) $\frac{n^2h^2}{4mL^2}$

Q10. Consider an oil drop from Millikan's oil drop experiment having diameter 1 μm confined between the plates separated by 10 μm. Density of oil is 900 kg/m^3. What is minimum energy of such an oil drop?

[Given : h = 6.63 × 10^{-34} Js]

(A) 2.4 × 10^{-42} J

(B) 1.2 × 10^{-42} J

(C) 3.6 × 10^{-42} J

(D) 4.8 × 10^{-42} J

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q11. In each situation of column–I a physical quantity related to orbiting electron in a hydrogen like atom is given. The terms 'Z' and 'n' given in column–II have usual meaning in Bohr's theory. Match the quantities in column–I with the terms they depend on in column–II :-

	Column-I		**Column-II**
(A)	Frequency of orbiting electron	(P)	Is directly proportional to Z^2
(B)	Angular momentum of orbiting electron	(Q)	Is directly proportional to n
(C)	Magnetic moment of orbiting electron	(R)	Is inversely proportional to n^3
(D)	The average current due to orbiting of electron	(S)	Is independent of Z
		(T)	None of these

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q12. Find the ratio of the area of orbit of first excited state of electron to the area of orbit of ground level for hydrogen atom.

Q13. When an electron jump from second orbit to ground state of hydrogen atom then calculate the wavelength of emitted photon.

SUBJECTIVE TYPE

Q14. A stationary hydrogen atom of mass M emits a photon corresponding to the longest wavelength of Balmer series. The recoil velocity acquired by the atom is-

(R = Rydberg constant and h = plank's constant)

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The velocity of an electron in single electron atom in an orbit.

(A) is independent of the atomic number of the element

(B) increases with increase in atomic number

(C) decreases with increase in atomic number

(D) increases with increase in quantum number

Q2. An energy of 24.6 eV is required to remove one of the electrons from a neutral helium atom. The energy (in eV) required to remove both the electrons form a neutral helium atom is -

(A) 38.2 (B) 49.2

(C) 51.8 (D) 79.0

Q3. If the atom $_{100}Fm^{257}$ follows the Bohr model and the radius of $_{100}Fm^{257}$ is n times the Bohr radius, then find n.

(A) 100 (B) 200

(C) 4 (D) 1/4

Q4. Let the potential energy of a hydrogen atom in the ground state be zero. Then its energy in the first excited state will be :

(A) 10.2eV (B) 13.6eV

(C) 23.8eV (D) 27.2eV

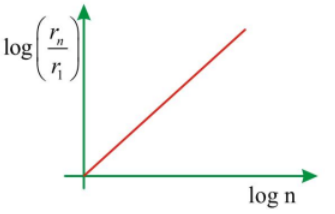
Q5. Ratio of difference of spacing between the energy levels with n = 3 and n = 4 and the spacing between the energy levels with n = 8 and n = 9 for a hydrogen like atom or ion is -

(A) 0.71 (B) 0.41

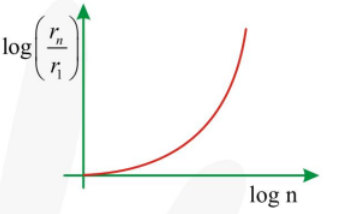
(C) 2.43 (D) 14.82

Q6. In hydrogen atom, the radius of nth Bohr orbit is r_n. The graph between $\log\left(\frac{r_n}{r_1}\right)$ and log n will be:

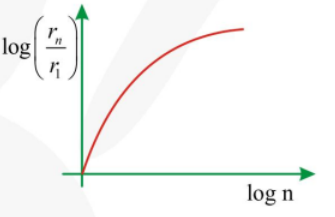
(A)



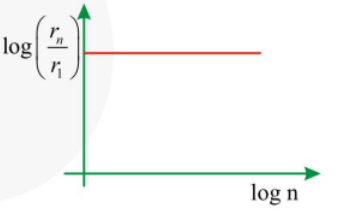
(B)



(C)



(D)



Q7. A photon collides with a stationary hydrogen atom in ground state inelastically. Energy of the colliding photon is 10.2 eV. After a time interval of the order of micro second another photon collides with same hydrogen atom inelastically with an energy of 15 eV. What will be observed by the detector?

(A) 2 photon of energy 10.2 eV.

(B) 2 photon of energy of 1.4 eV.

(C) One photon of energy 10.2 eV and an electron of energy 1.4 eV.

(D) One photon of energy 10.2 eV and another photon of energy 1.4 eV.

Q8. If $\lambda = c_2\left[\frac{n^2}{n^2-2^2}\right]$ for balmer series, what is the value of c_2 ?

(A) $\frac{4}{R_H}$ (B) $\frac{2}{R_H}$

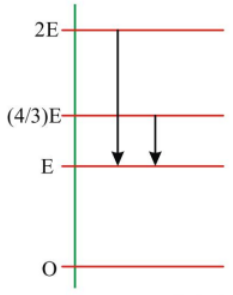
(C) $2R_H$ (D) $4R_H$

Q9. The largest wavelength in the lymen series of the hydrogen spectrum is 122 nm. The smallest wavelength in the paschen series of the hydrogen spectrum (to the nearest integer) is :

(A) 802 nm (B) 823 nm

(C) 1882 nm (D) 1648 nm

Q10. The figure shows energy levels of a certain atom, when the system moves from level 2E to E, a photon of wavelength λ is emitted. The wavelength of photon produced during its transition from level 4/3 E to E level is :



(A) 3λ (B) 3/4λ

(C) λ/4 (D) 2λ

Q11. Calculate the energy of a photon of sodium light of wavelength 2×10^{-16} m ?

(A) 9.9×10^{-10} J (B) 6.6×10^{-11} J

(C) 9.9×10^{-12} J (D) 4.8×10^{-11} J

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Let A_n be the area enclosed by the nth orbit in a hydrogen atom.

The graph of $\ln(A_n / A_1)$ against $\ln(n)$:

(A) will pass through the origin

(B) will be certain points lying on a straight line with slope 4

(C) will be a monotonically increasing nonlinear curve

(D) will be a circle

Q2. The electrons in a hydrogen atom make a transition $n_1 \rightarrow n_2$ are the principal quantum numbers of two states. Assume the Bohr's model to be valid. The time period of the electron in the initial state is eight times that in the final state. The possible values of n_1 and n_2 are :

(A) $n_1 = 4, n_2 = 2$ (B) $n_1 = 8, n_2 = 2$

(C) $n_1 = 8, n_2 = 1$ (D) $n_1 = 6, n_2 = 3$

Q3. In Bohr's model of the hydrogen atom, let R. V. T, and E represent the radius of the orbit, speed of the electron, time period of revolution of electron, and the total energy of the electron, respectively. The quantities proportional to the quantum number *n* are :

(A) VR (B) RE

(C) $\frac{T}{R}$ (D) $\frac{V}{E}$

Q4. Imagine an atom made up of proton and a hypothetical particle of double the mass of the electron but having the same charge as the electron. Apply the Bohr atom model and consider all possible transitions of this hypothetical particle to the first excited level. The longest wavelength photon that will be emitted has wavelength λ (given in terms of the Rydberg constant R for hydrogen atom) equal to :

(A) 9/5 R (B) 36/5R

(C) 18/5R (D) 4/R

Q5. A hydrogen atom and a Li^{2+} ion are both in the second excited state. If ℓ_H and ℓ_{Li} are their respective electronic angular momenta and E_H and E_{Li} their respective energies, then :

(A) $\ell_H > \ell_{Li}$ and $|E_H| > |E_{Li}|$

(B) $\ell_H = \ell_{Li}$ and $|E_H| < |E_{Li}|$

(C) $\ell_H = \ell_{Li}$ and $|E_H| > |E_{Li}|$

(D) $\ell_H < \ell_{Li}$ and $|E_H| < |E_{Li}|$

Q6. The electron in a hydrogen atom makes a transition from an excited state to the ground state. Which of the following statement is true?

(A) Its kinetic energy increases and its potential and total energy decreases

(B) Its kinetic energy decreases, potential energy increases and its total energy remains the same

(C) Its kinetic and total energy decreases and its potential energy increases

(D) Its kinetic, potential and total energy decrease

Q7. In the hydrogen atom, if the reference level of potential energy is assumed to be zero at the ground state level. Choose the incorrect statement.

(A) The total energy of the shell increases with increase in the value of n.

(B) The total energy of the shell decrease with increase in the value of n.

(C) The difference in total energy of any two shells remains the same.

(D) The total energy at the ground state becomes 13.6 eV.

Q8. When a hydrogen atom is excited from ground state to first excited state then :

(A) Its kinetic energy increases by 10.2 eV.

(B) Its kinetic energy decreases by 10.2 eV.

(C) Its potential energy increases by 20.4 eV.

(D) Its angular momentum increases by 1.05×10^{-34} J–s.

Q9. Suppose the potential energy between electron and proton at a distance r is given by $-\frac{Ke^2}{3r^3}$.

Application of Bohr's theory to hydrogen atom in this case shows that :

(A) energy in the nth orbit is proportional to n^6

(B) energy is proportional to m^{-3} (m = mass of electron)

(C) energy in the nth orbit is proportional to n^{-2}

(D) energy is proportional to m^3 (m = mass of electron)

Q10. Hydrogen atoms absorb radiation of wavelength λ_0 and consequently emit radiations of 6 different wavelengths of which two wavelengths are shorter than λ_0. Then -

(A) The final excited state of the atoms is $n = 4$

(B) The initial state of the atoms may be $n = 2$

(C) The initial state of the atoms may be $n = 3$

(D) There are three transitions belonging to Lyman series

Q11. The electron in a hydrogen atom jumps back from an excited state to ground state, by emitting a photon of wavelength $\lambda_0 = \frac{16}{15R}$, where R is Rydberg's constant. In place of emitting one photon, the electron could come back to ground state by :

(A) Emitting 3 photons of wavelengths λ_1, λ_2 and λ_3 such that $\frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{1}{\lambda_3} = \frac{15R}{16}$

(B) Emitting 2 photons of wavelength λ_1 and λ_2 such that $\frac{1}{\lambda_1} + \frac{1}{\lambda_2} = \frac{15R}{16}$

(C) Emitting 2 photons of wavelength λ_1 and λ_2 such that $\lambda_1 + \lambda_2 = \frac{16}{15R}$

(D) Emitting 3 photons of wavelength λ_1, λ_2 and λ_3 such that $\lambda_1+\lambda_2+\lambda_3 = \frac{16}{15R}$

Q12. In a hydrogen atom, the electron is in n^{th} excited state. It may come down to second excited state by emitting ten different wavelengths. What is the value of n ?

(A) 6 (B) 7

(C) 8 (D) 5

Q13. Monochromatic radiation of wavelength λ is incident on a hydrogen sample containing in ground state. Hydrogen atoms absorb the light and subsequently emit radiations of ten different wavelengths. The value of λ is -

(A) 95 nm (B) 103 nm

(C) 273 nm (D) 88 nm

Q14. The electron in hydrogen atom in a sample is in n^{th} excited state, then the number of different spectrum lines obtained in its emission spectrum will be :

(A) $1 + 2 + 3 + ... + (n - 1)$

(B) $1 + 2 + 3 + ... + (n)$

(C) $1 + 2 + 3 + ... + (n + 1)$

(D) $1 \times 2 \times 3 \times .....\times (n - 1)$

Q15. The electric potential between a proton and an electron is given by $V = V_0 \ell n \frac{r}{r_0}$, where r_0 is a constant. Assuming Bohr's model to be applicable, write variation of r_n with n, n being the principal quantum number :–

(A) $r_n \propto n$ (B) $r_n \propto \frac{1}{n}$

(C) $r_n \propto n^2$ (D) $r_n \propto \frac{1}{n^2}$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question Nos. 16 to 18

An electron is orbiting in a circular orbit of radius r under the influence of a constant strong magnetic field of strength B. Assuming that Bohr's postulate regarding the quantisation of angular momentum holds good for this electron, find :

Q16. The allowed values of the radius 'r' of the orbit $\left(\text{here } a_0 = \sqrt{\frac{h}{2\pi eB}}\right)$.

(A) $r = \sqrt{n}a_0$ (B) $r = \frac{1}{2}\sqrt{n}a_0$

(C) $r = 2\sqrt{n}a_0$ (D) None of these

Q17. The time period of revolution of electron in n^{th} orbit is :-

(A) $\frac{3\pi m}{eB}$ (B) $\frac{\pi m}{eB}$

(C) $\frac{\pi m}{2eB}$ (D) $\frac{2\pi m}{eB}$

Q18. The total magnetic flux due to the magnetic field B passing through the nth orbit.

(A) $\phi = \frac{2nh}{e}$ (B) $\phi = \frac{nh}{e}$

(C) $\phi = \frac{nh}{2e}$ (D) $\phi = \frac{4nh}{e}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q19. When we write expression for energy of electron in n^{th} orbit of helium ion (He^+) we take zero potential energy for $n = \infty$, but the potential energy depends on reference. If we take total energy of atom for n = 1 orbit as zero then

Column–I	**Column–II**
(A) Total energy of electron in n = 2	(P) 54.4 eV
(B) Ionization energy from ground state	(Q) 40.8 eV
(C) Energy required to exit electron from n = 1 to n = 2	(R) depends on reference level
(D) Negative of potential energy of electron in n = 1	(S) independent of reference level.
	(T) 70.3 eV

Q20. The energy, the magnitude of linear momentum, magnitude of angular momentum and orbital radius of an electron in a hydrogen atom corresponding to the quantum number n are E, p, L and r respectively. Then according to Bohr's theory of hydrogen atom, match the expressions in column-I with statement in column-II.

Column-I	**Column-II**
(A) Epr	(P) is independent of n.
(B) $\frac{p}{E}$	(Q) is directly proportional to n
(C) Er	(R) is inversely proportional to n.
(D) pr	(S) is directly proportional to L.

NUMERICAL TYPE

Q21. An electron in the nth excited state in a hydrogen atom comes down to the first excited state by emitting 10 different wavelengths. Find the value of n (an integer).

Q22. Find the recoil speed (appoximately in m/sec) when a hydrogen atom emits a photon during the transition from n = 5 to n = 1.

SUBJECTIVE TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q23. Find the longest and shortest wavelength when a hydrogen atom in the ground state is excited by radiations of wavelength 975 Å.

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	A	A	D	C	C	D	C	D	D
11	**12**	**13**	**14**	**15**	**16**	**17**			
D	C	D	A	C	B	B			

Ex-1A

1. A,C,D **2.** B **3.** A,B,C,D **4.** A,C,D **5.** C

6. A,C,D **7.** B **8.** B **9.** C **10.** B

11. (A) $\rightarrow$(P,R) ; (B) $\rightarrow$(Q,S) ; (C) $\rightarrow$(Q, S) ; (D)$\rightarrow$(P,R)

12. 16 : 1 **13.** 121 nm. **14.** $\frac{Rh}{M}\left(\frac{5}{36}\right)$

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	D	D	C	B	A	C	A	B	A
11									
A									

Ex-2A

1. A,D **2.** A,D **3.** A,C,D **4.** C **5.** B

6. A **7.** B **8.** B,C,D **9.** A,B **10.** A,B,D

11. A,B **12.** A **13.** A **14.** B **15.** A

16. A **17.** D **18.** C **19.** (A)-QR, (B)-PS, (C)-QS, (D)-PR

20. (A)-R, (B)-Q,S (C)-P, (D)-Q,S **21.** 6 **22.** 4

23. $\lambda_{max} = 18787.8$ Å, $\lambda_{min} = 975$ Å